

Robot Arm Electrical Report

Motor Driver PCB, SCADA/PID Robot Arm Simulation, and MATLAB/Multisim Validation

Converted from project report for portfolio evidence.

Abstract

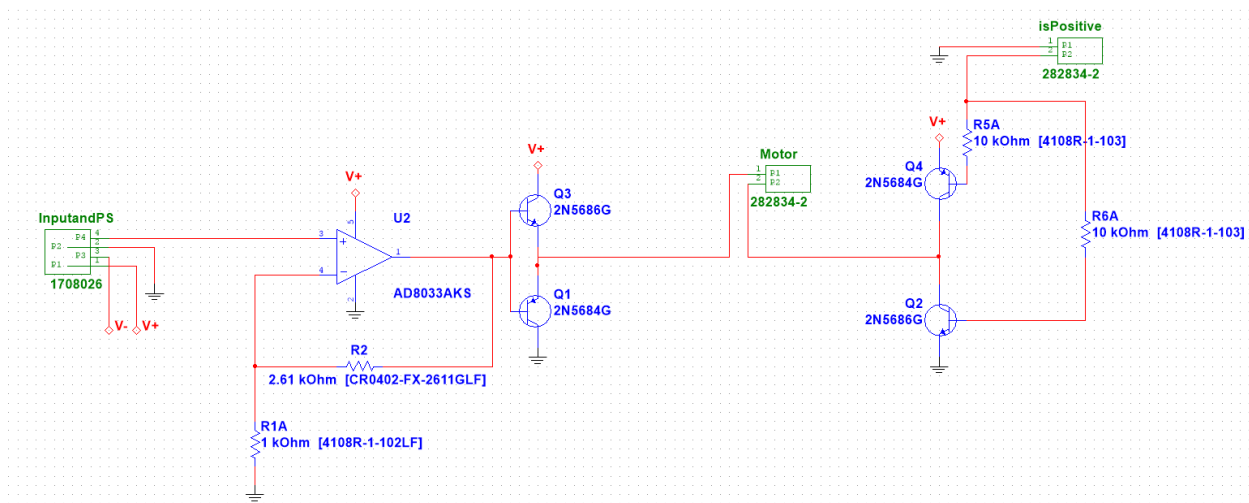
This report details the design and production of a motor driver PCB for use in a quality-control robot arm. The PCB consists of components to power 4 motors to be used in the quality-control robot. The design features 4 separate circuits integrated together, each consisting of an input for the power supply and Arduino (used to control the system) as well as outputs for the motor to be powered. The report begins with a Preface describing tests performed to select an optimal circuit. Section A describes the Multisim circuit drawing and design. Section B describes the Multisim step response. Section C describes the linear MATLAB model used. Section D describes the MATLAB step response. Section E describes the PCB design and result in Ultiboard. Finally, Section F summarizes the cost of the components used in the circuit.

Preface

Motor Driver Circuit Selection:

For each motor, a total of 2 circuits (Emitter-follower and H-bridge) were designed and subsequently compared. Analysis finds the H-bridge design to be better for a multitude of reasons:

The H-Bridge requires only 4 high-power transistors in order to allow for negative voltage. The Emitter-follower design would require 2 op-amps with an inverter. Moreover, it would require 3 more MOSFETs to control the signal going into the inverter (selecting whether to bypass the inverter or not to control the motor direction). This design comparison between the Emitter-follower and H-Bridge is shown below in Fig. 1.



The design above would require the use of a signal to control whether or not to go through the inverter (for the motor to go left or right). Hence, a MOSFET switch is added as described below in Fig. 2:

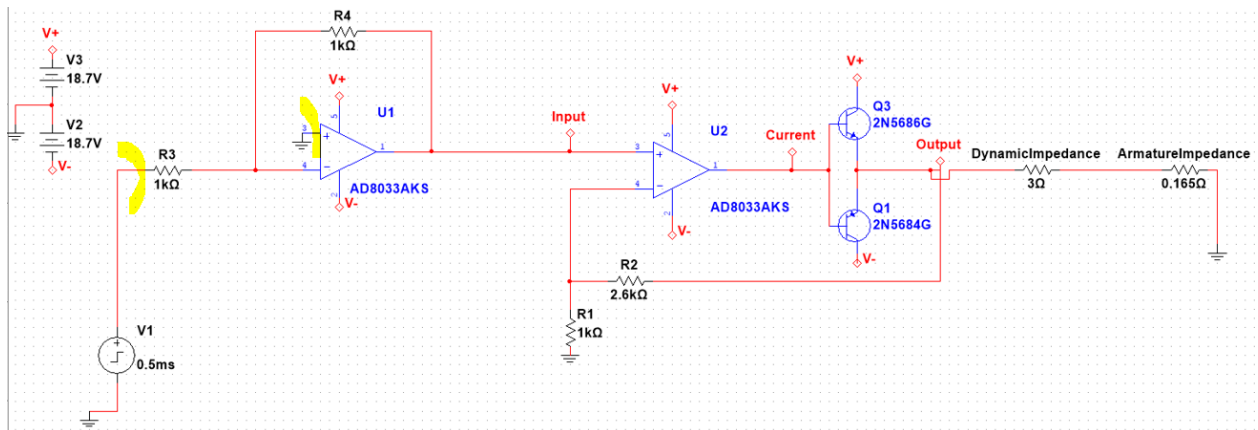
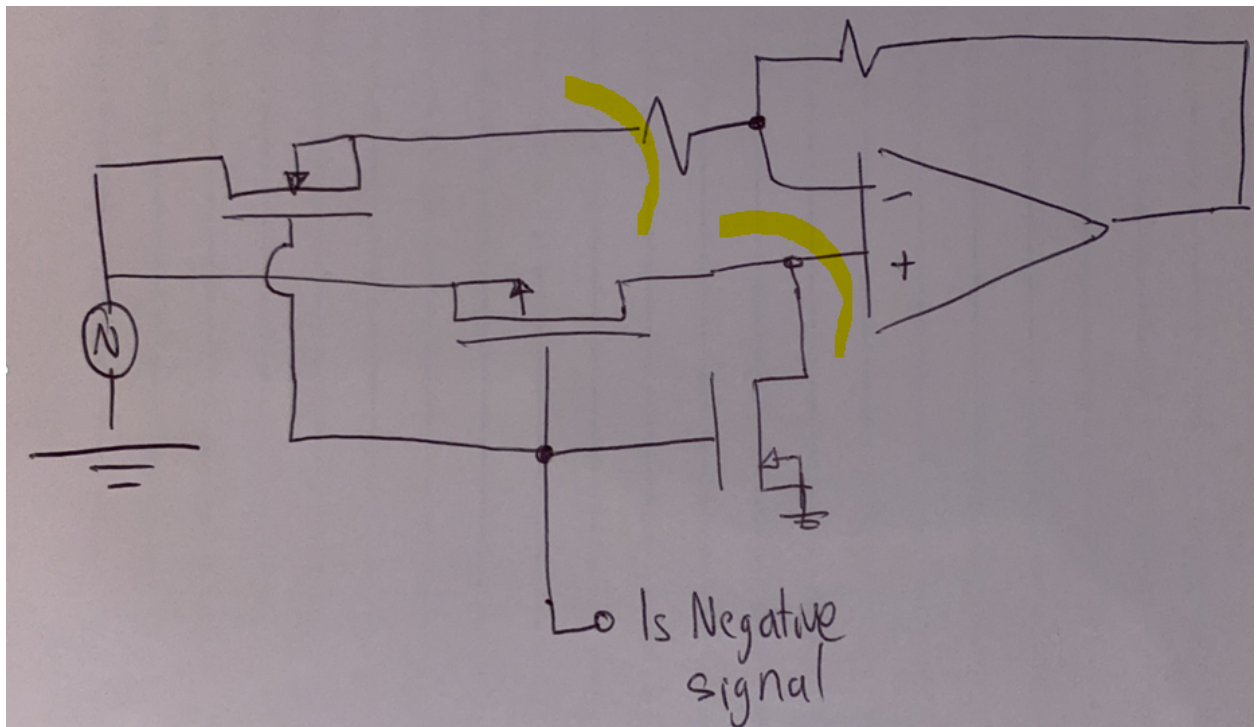


Figure 1: Emitter-follower Design with Inverter



In the sketch above, the hand-drawn circuit would be integrated where the yellow high-lighter marking is added. This would allow for a control signal to select if voltage would pass through the inverter to control the motor direction.

Logic for controlling motor with Is Negative signal

For reasons explained above, the H-bridge design was selected as shown in Fig 3:

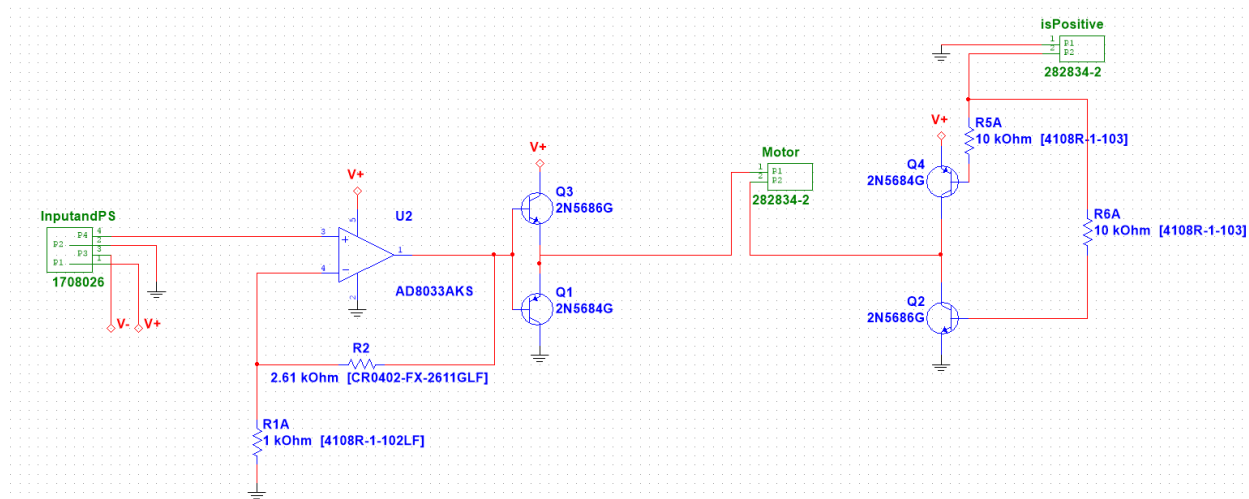


Figure 2: Emitter-follower Design with Inverter and Is Negative Signal

Motor modeling for response:

Using the DCX32L motor:

All motors including the DCX32L have dynamic impedance. For instance, when moving full speed under no load conditions is $18/.234A=76.9$ Ohms [Armature + Dynamic impedance] (where .234A is no load current draw). When this motor starts/stalls, it will see stall current (very high 100+A) and 0.165 Ohms [Dynamic =0, Total impedance = armature]. Thus, a driver circuit for this motor would need to have a transistor that can hold 100+A. This would require a Darlington set-up with high current to the base. It is more accurate to model the motor with armature impedance and dynamic impedance according to the design constraints. This way can the BJT will go to saturation and allow for ~18V for motor. This method of modeling the motor impedance was confirmed with a TA during the lab section as a more accurate method.

An example calculation (in this case the one used for the DCX32L motor) is shown below:

Dynamic Impedance under load calculation done in reverse:

Calculating how much power is required for motor to produce:

Upper-bound (will not go over no load speed with load – worst case scenario):

No load speed (in rad/s) = $8630/(60)*2\pi=903$ – using worst case scenario

$903*87.87mNm=79W$ – here 87.87mNm is torque of arm.

Finding correct motor dynamic impedance battery/source will see under constraint of arm:

Modeling the total impedance of motor as 3 [dynamic impedance] +0.165 Ohms. Thus: $18=3.165*I=5.6872A$ for the motor. $P=5.6872*18=108W$... assuming 80% efficiency, arrive at: 81.89W. 81W is enough for the motor to move the arm.

Note: The alternative (armature resistance only without dynamic impedance) to this method does not work well in simulation with the DCX32L. Without including dynamic impedance for this motor, the source current to the motor would have to be extremely high which is unrealistic. This would require the circuit to use Darlington BJTs to handle this extremely high current. Moreover, it should be noted that this method is still an estimation. It is not 100% accurate the load due to the load only representing steady-state conditions. It does not take into account current spike at start which might be too high for motor.

Each motor circuit was designed with an Emitter follower design and a H-bridge design for comparison. An H-Bridge design was selected for reasons explained in the preface.

DCX14L@18V:

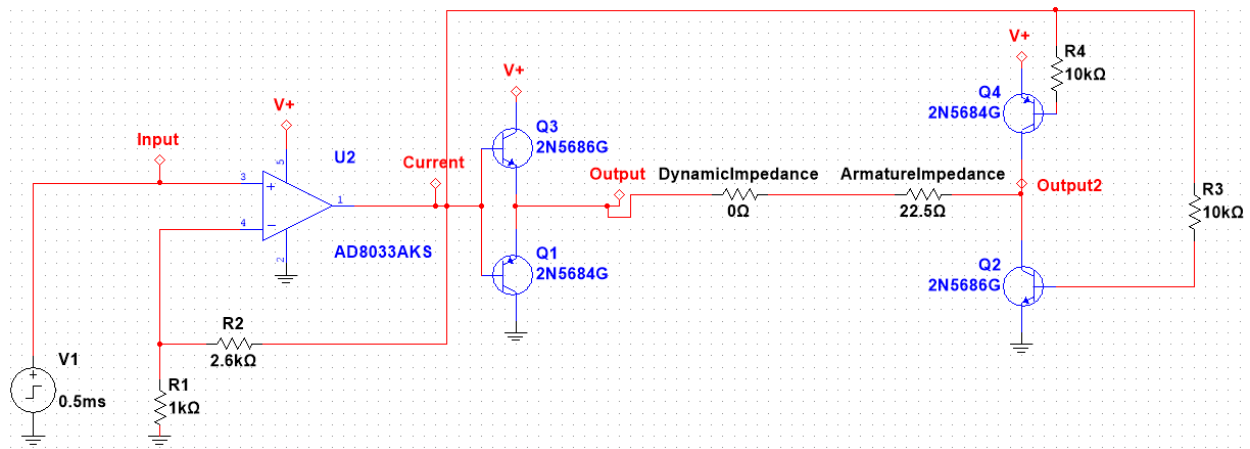


Figure 5: H-bridge motor driver circuit which was used in the project

DCX10L@6V:

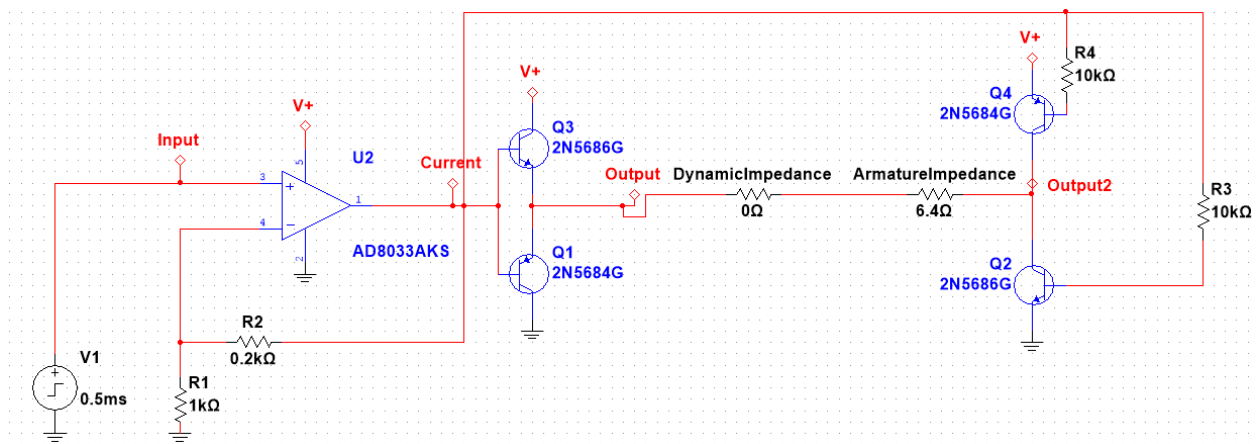


Figure 6: H-bridge motor driver circuit which was used in the project

B) Multisim Step Response

The circuits shown previously were all subject to transient simulation in Multisim to confirm the amplifiers work within spec. The design required an amplifier for a 5V (Arduino) input to either 18V (DCX32/22/14L) or 6V (DCX10L). The transient responses are shown in Fig. 8, 9, 10, 11 and 12.

DCX32L@18V:

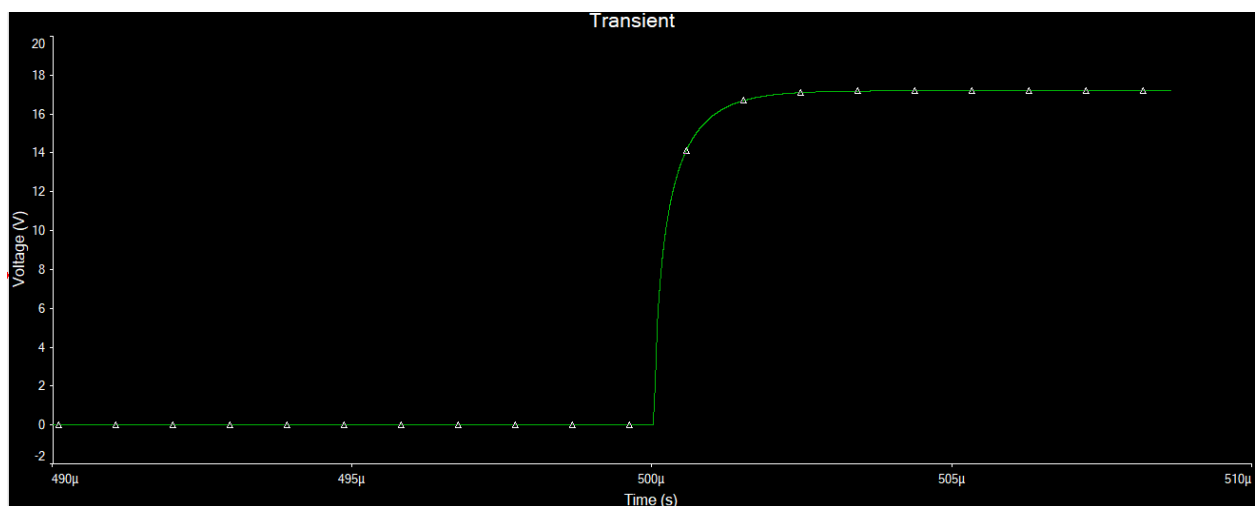
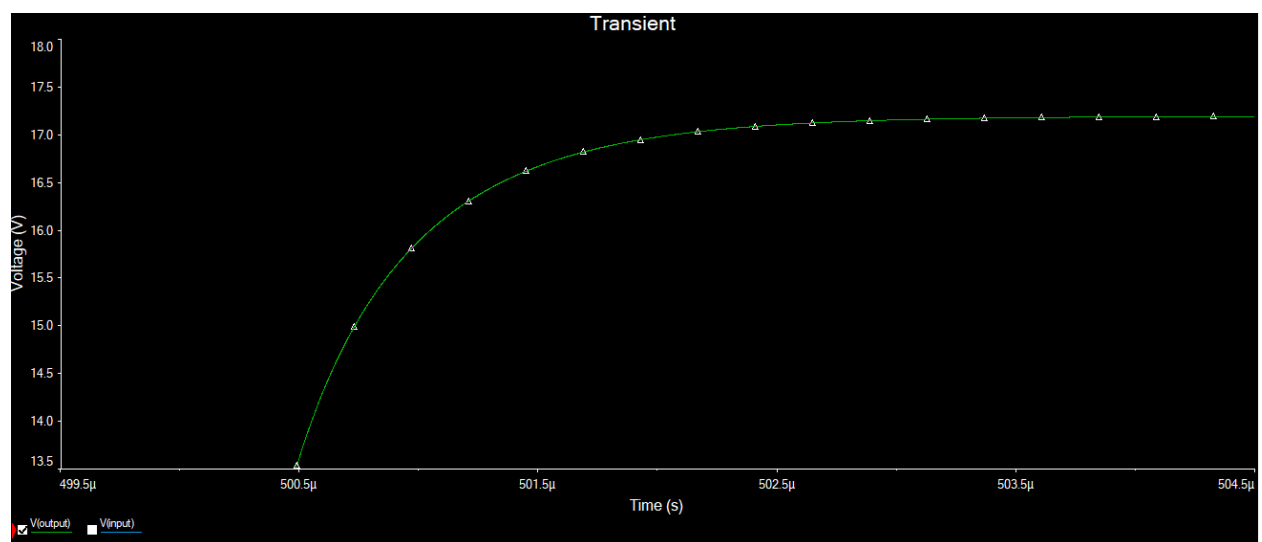


Figure 7: H-bridge motor driver circuit which was used in the project



DCX22L@18V:

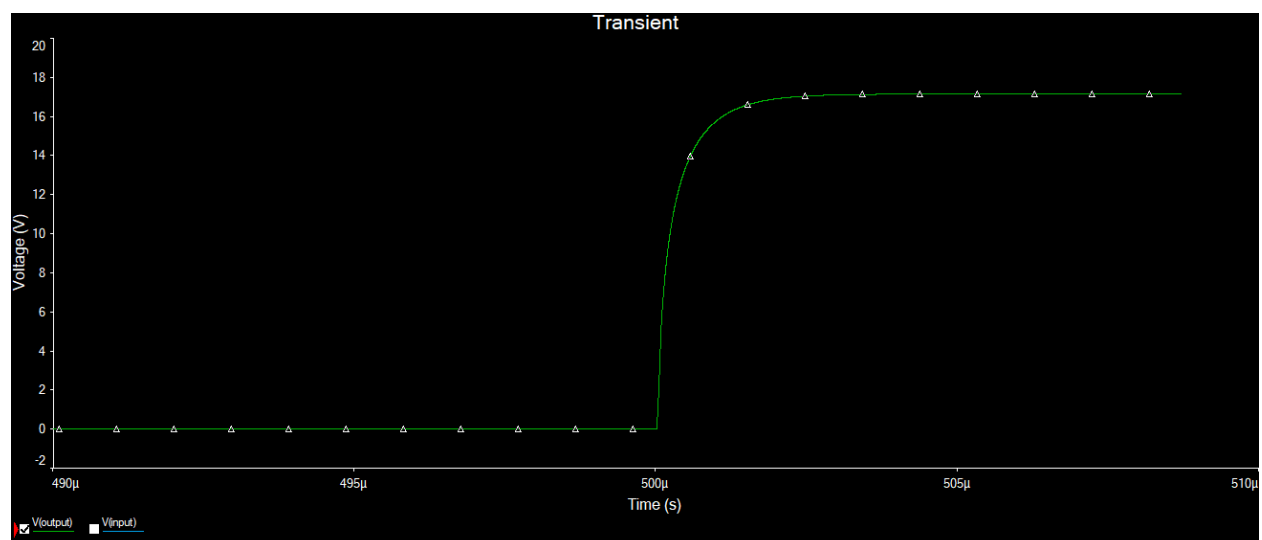
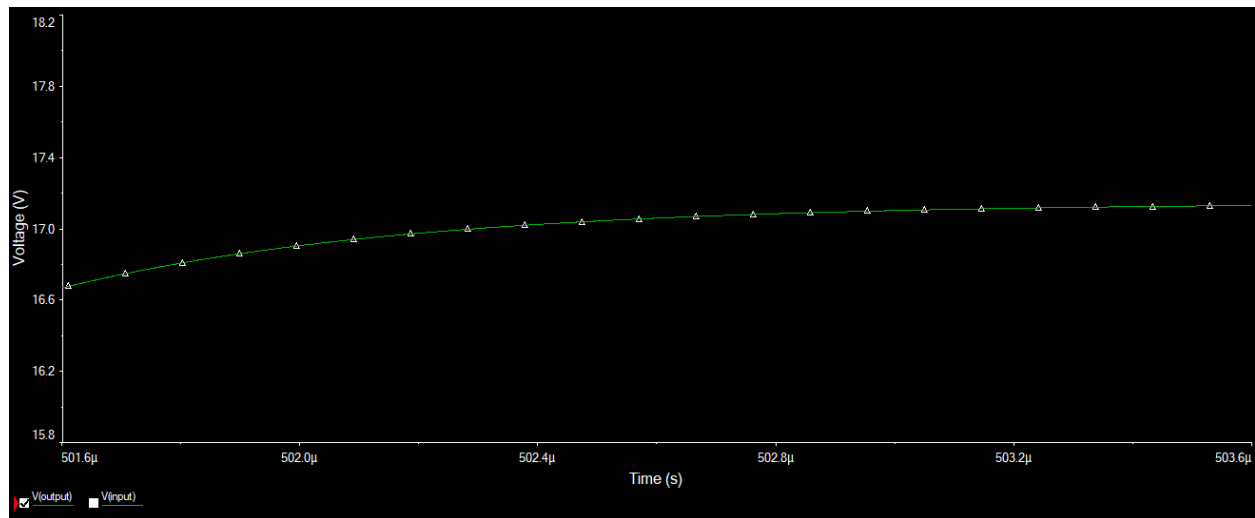


Figure 8: MultiSim Transient Step response with zoom below



DCX14L@18V:

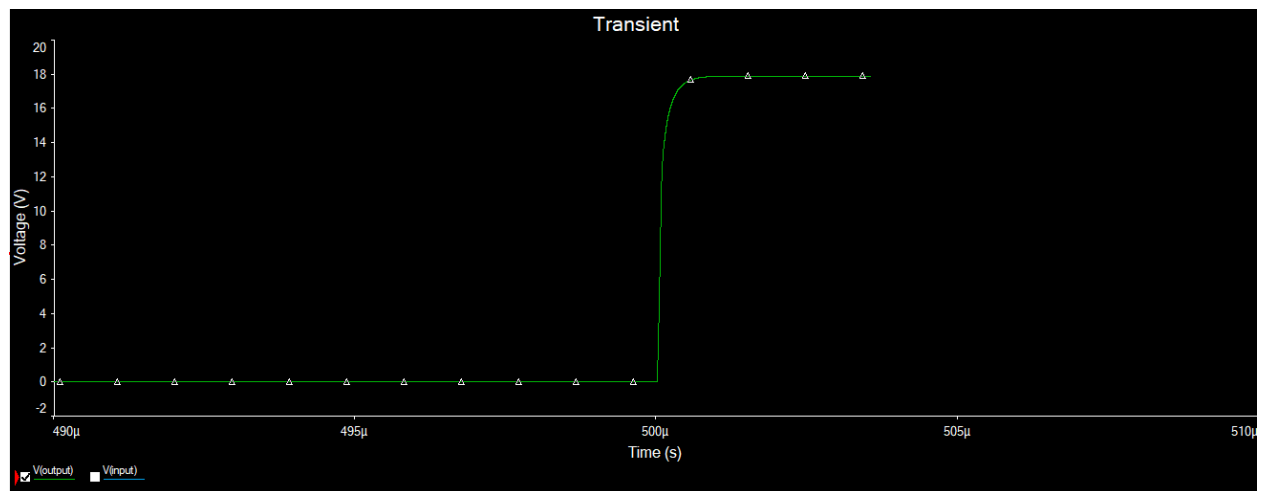
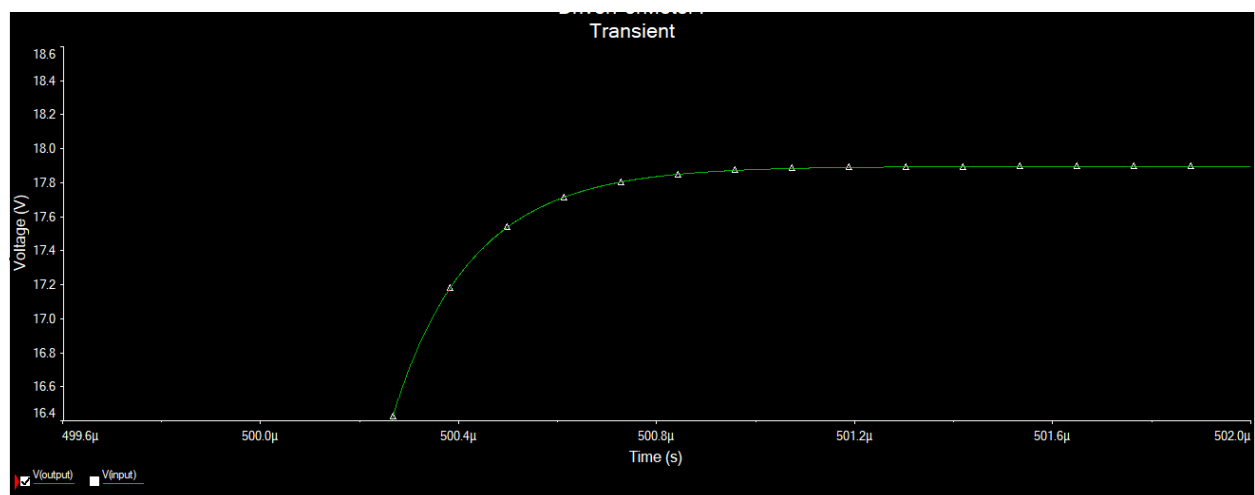


Figure 9: MultiSim Transient Step response with zoom below



DCX10L@6V:

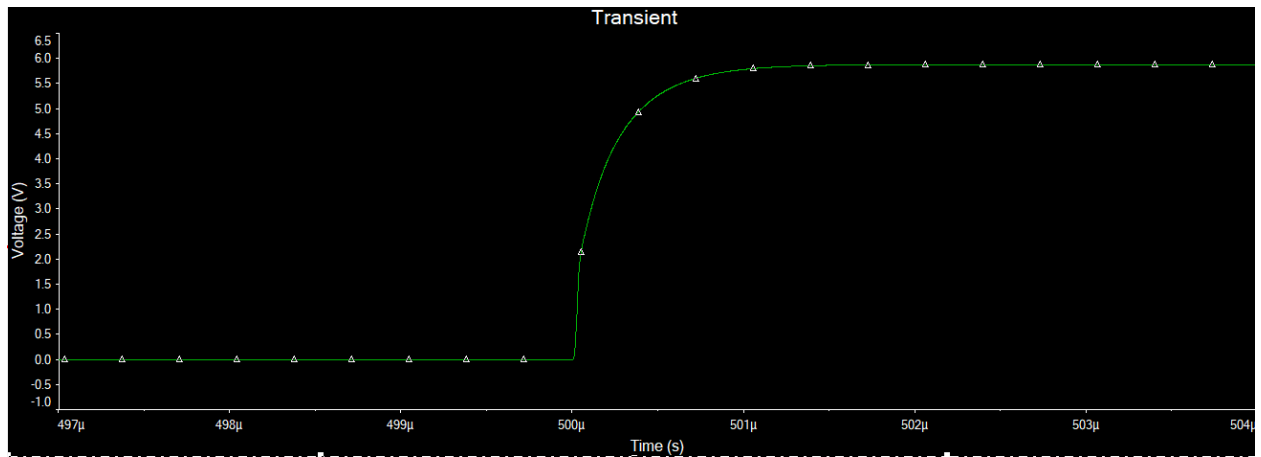


Figure 10: MultiSim Transient Step response with zoom below

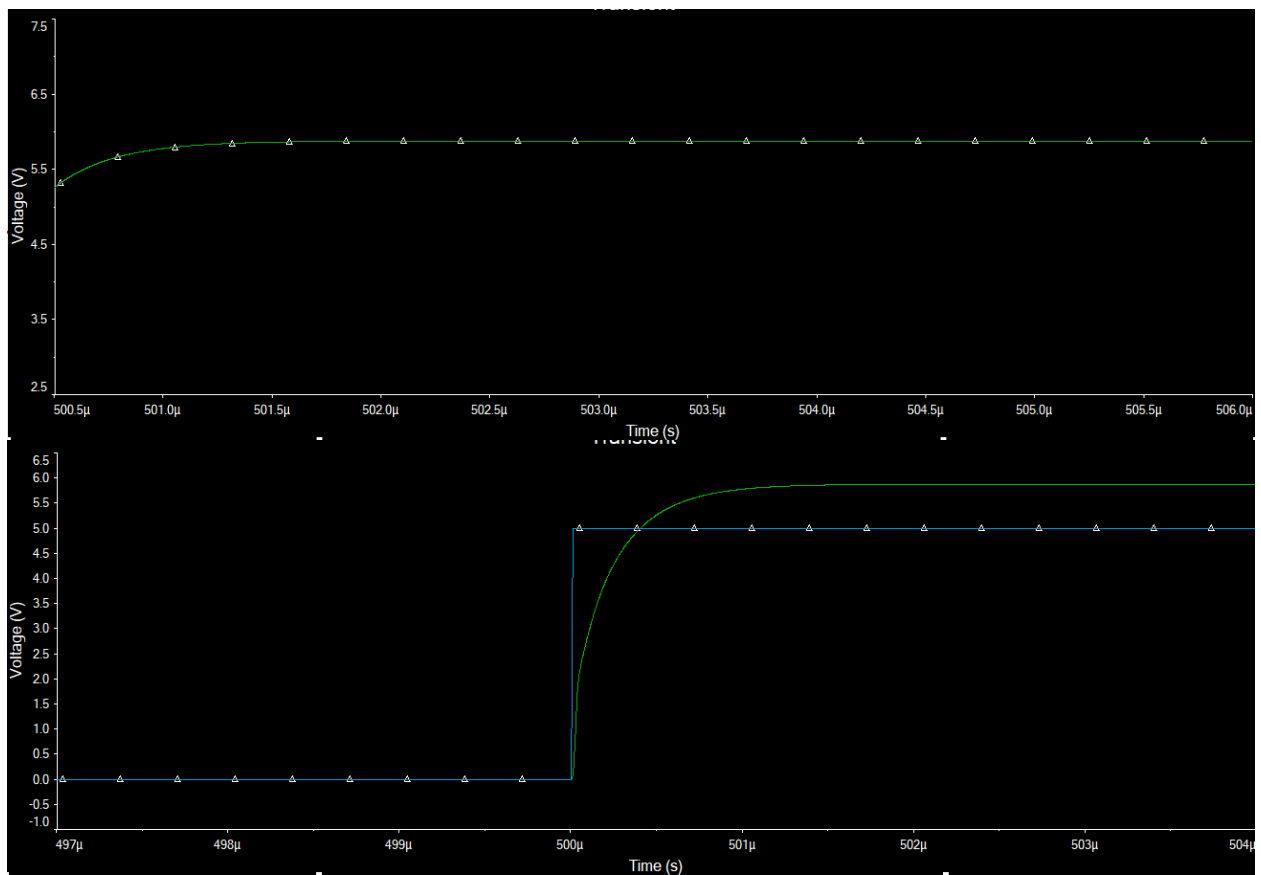


Figure 11: MultiSim Transient Step response with zoom below

C) Linear MATLAB Model

From the datasheet shown in Appendix A, the motor parameters shown below are calculated. The calculations are done in MATLAB with the formulas shown in Fig. 13.

DCX32L

DCX22L

DCX14L

DCX10L

The code used in MATLAB to simulate the motors is shown below. Here the data is filled in for the DCX32L.

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Initializes variables for the Simulink model of an electric motor %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Start with a clean slate
% clear all;
%Amplifier from Multisim:
sim_data = iddata(OutV, InV, 1e-9);
Motor AMP TF = -tfest(sim_data, 1, 0)
% Motor parameters:
R = 0.165; % Armature resistance 15 Terminal resistance ohm
L = 0.053 / 1000; % Armature inductance 11 Terminal inductance mH -> H

no_loadcurrent = 234 / 1000; %3 No load current mA -> A
torqueconstant = 19.5 / 1000; %12 Torque Constant mNm/A -> Nm/A
no_loadspeed = 8630 * 2*pi/60; %2 No load speed rpm -> rad/sec
speedConstant = 490;

% Mechanical
J = 75.9 * 10^(-7); % Armature inertia 16_Motor Interia gcm2 -> kgm2
B = (no_loadcurrent*torqueconstant) / (no_loadspeed); % Armature damping

% Bridge
Km = torqueconstant; % Motor constant 12 Torque Constant mNm/A -> Nm/A
Kb = 1/(speedConstant * 2*pi/60); % Back-EMF constant 13_ Speed
constant rpm/V -> V/(rad/sec)->V*sec/rad

% Display some results
G = tf(Km, [L*s (L*B + R*J) R*B]);
H = tf(Kb, 1);
motVor TF = feedback(G,H)

```

Figure 12: MultiSim Transient Step response showing both input and output (input of 5V and output of ~6V in this case) Input (blue curve) is the same for each motor, only output changes.

The MATLAB code above works together with the following Simulink model to complete the linear motor model shown in Fig. 14:

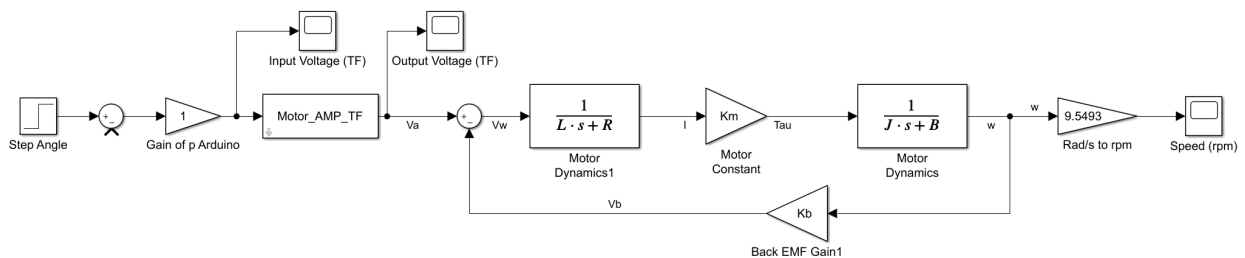


Figure 13: Simulink motor model (open loop)

D) MATLAB Step Response

The linear motor model made in Simulink produces the following step responses. When compared to the no-load speed for each motor, it is apparent the model is functioning as expected (low margin of error).

DCX32L @18V

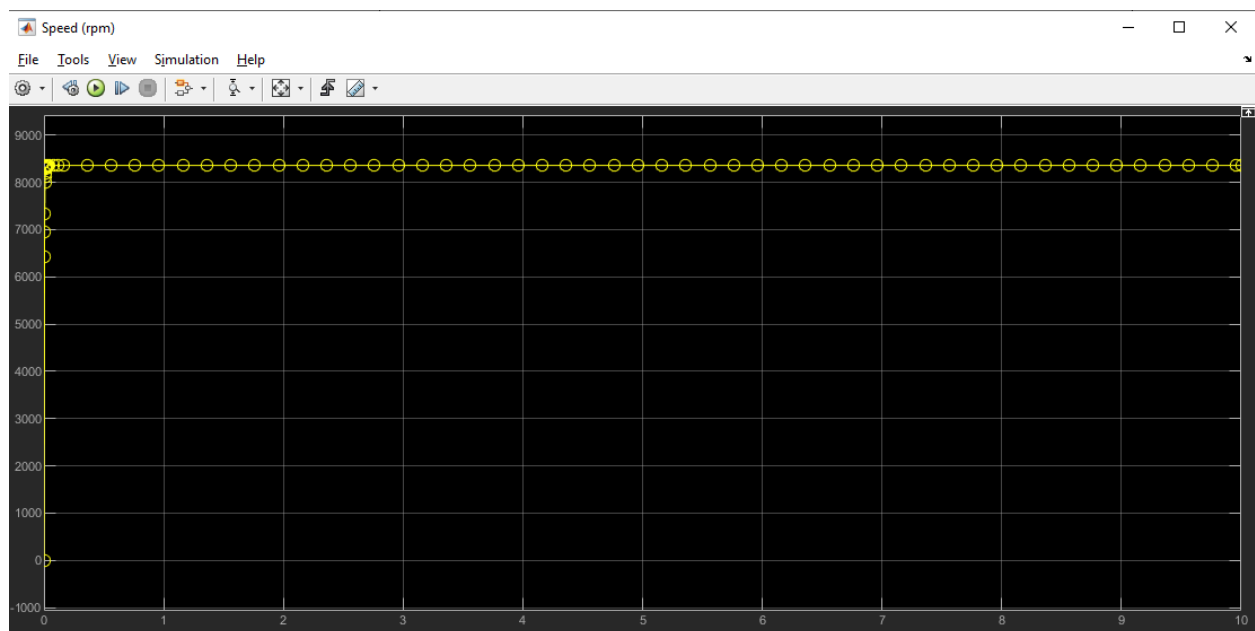


Figure 14: Simulink motor model (open loop)

DCX22L @18V

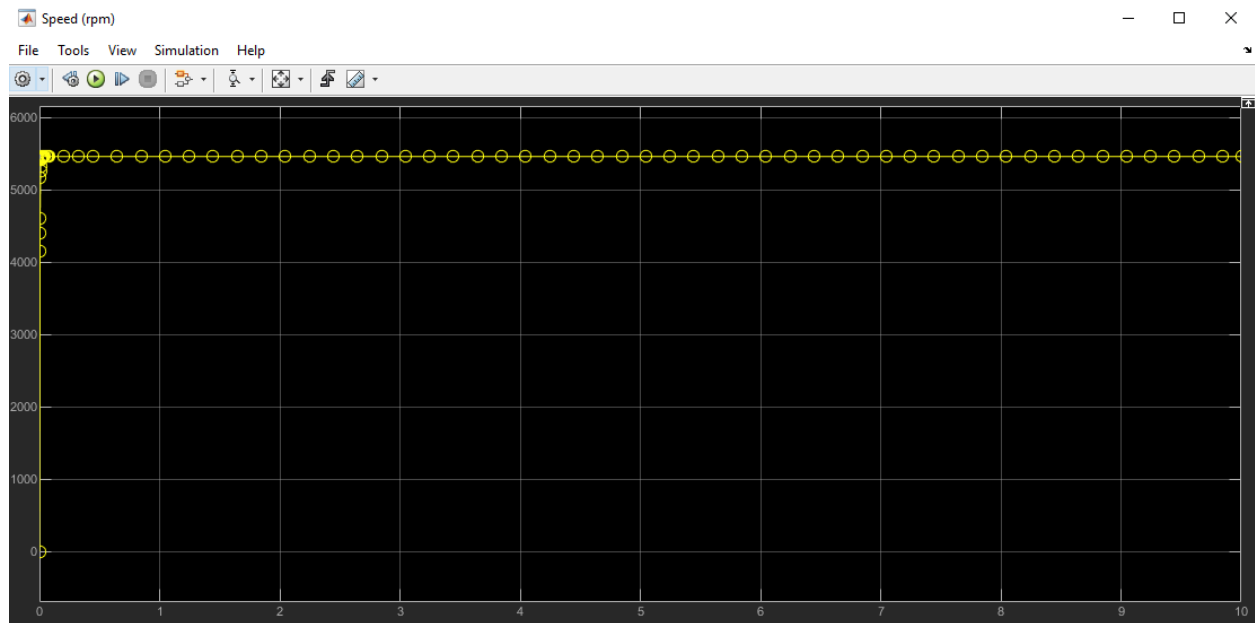


Figure 15: DCX32L Step Response in (speed in RPM)

DCX14L @18V



Figure 16: DCX22L Step Response in (speed in RPM)

DCX10L @6V

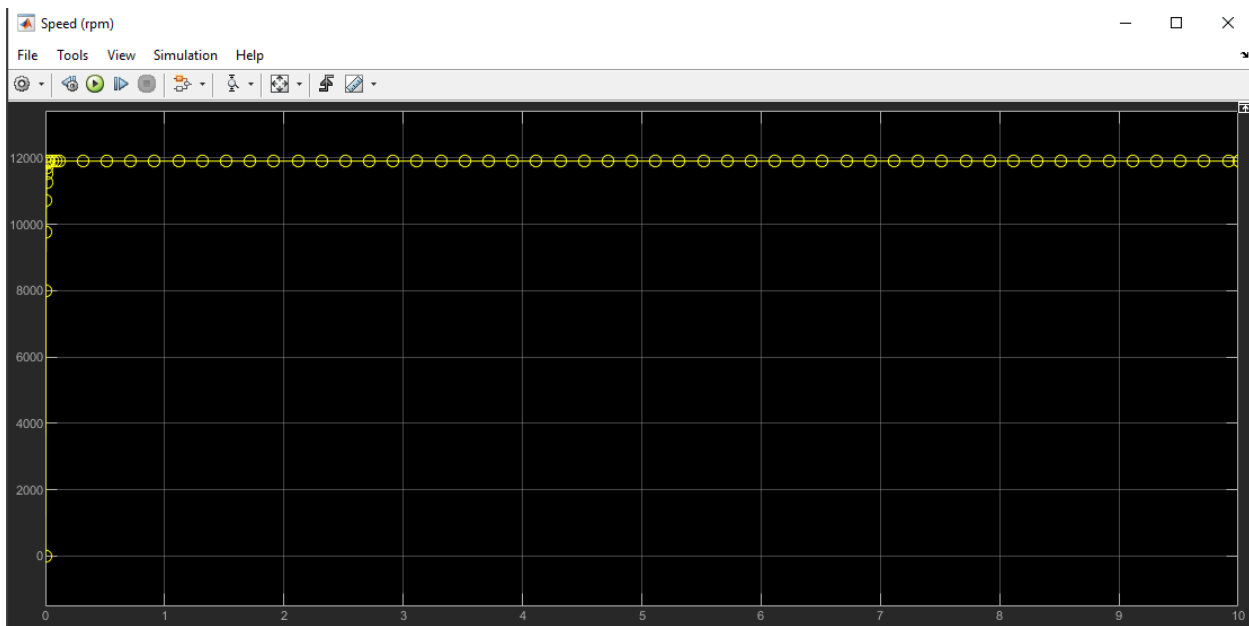


Figure 17: DCX14L Step Response in (Speed in RPM)

Amplifier Response (5V to ~18V)

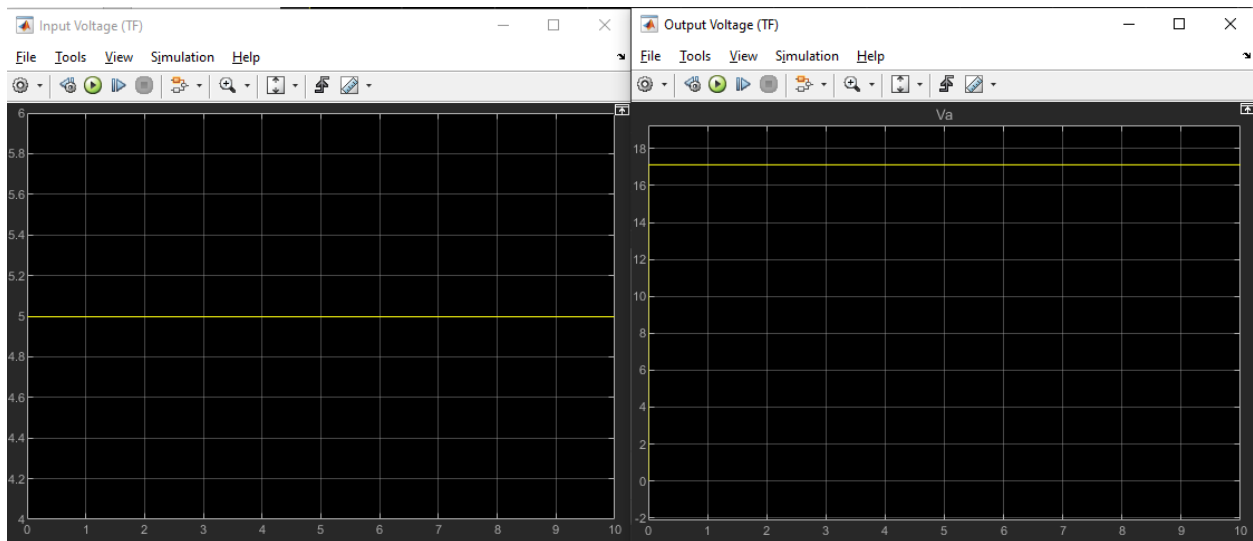


Figure 18: DCX10L Step Response in (speed in RPM)

Amplifier Response (5V to ~6V)

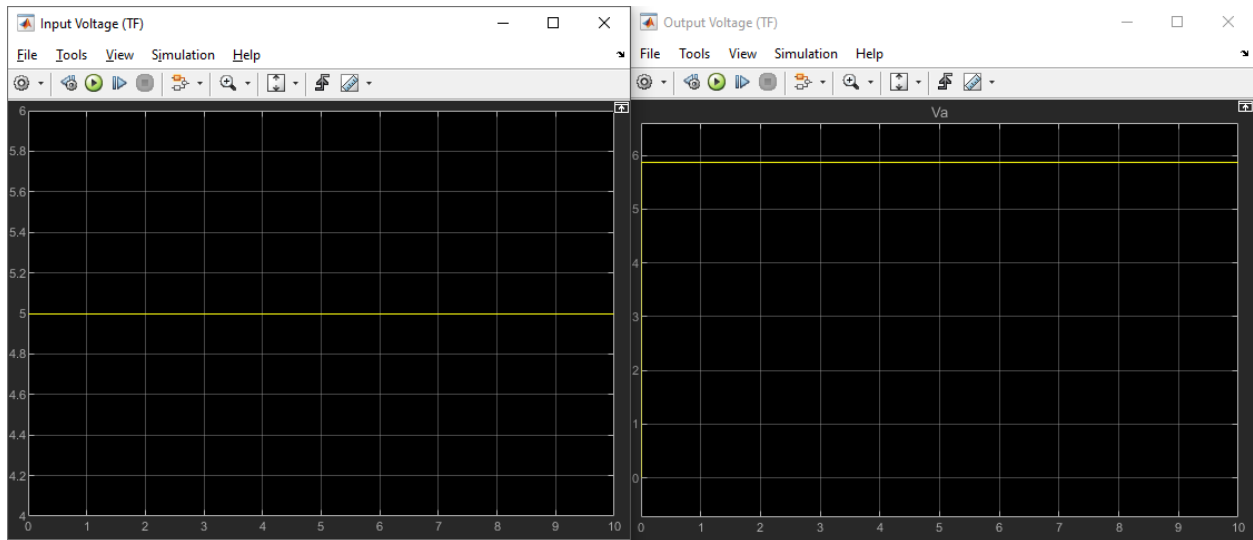


Figure 19: Amplifier Step response for DCX32L/22L/14L

E) PCB Layout

The PCB was laid out to maximize efficiency. The overall circuit shown in Fig. 23 consists of 3 copies of the circuit in Fig. 21 and one of Fig. 22 (different feedback resistors). As there are 4 motors, each motor has a dedicated driver circuit and they are just connected together on one PCB. The inputs are located on one side and the outputs on the other for easy-access. Moreover, the BJTs are stacked vertically for each motor's circuit, decreasing the amount of trace distance (maximizing design efficiency thus decreasing complexity and overall cost).

Circuits for H-Bridge Circuit:

All motors except for DCX10L:

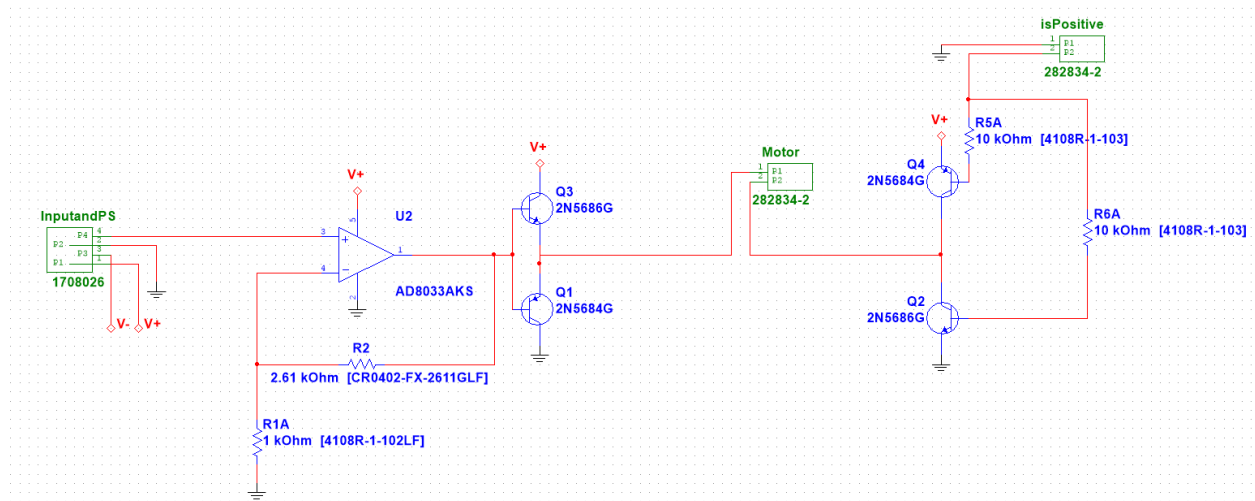


Figure 20: Amplifier Step response for DCX10L

DCX10L (Different resistor required for feedback):

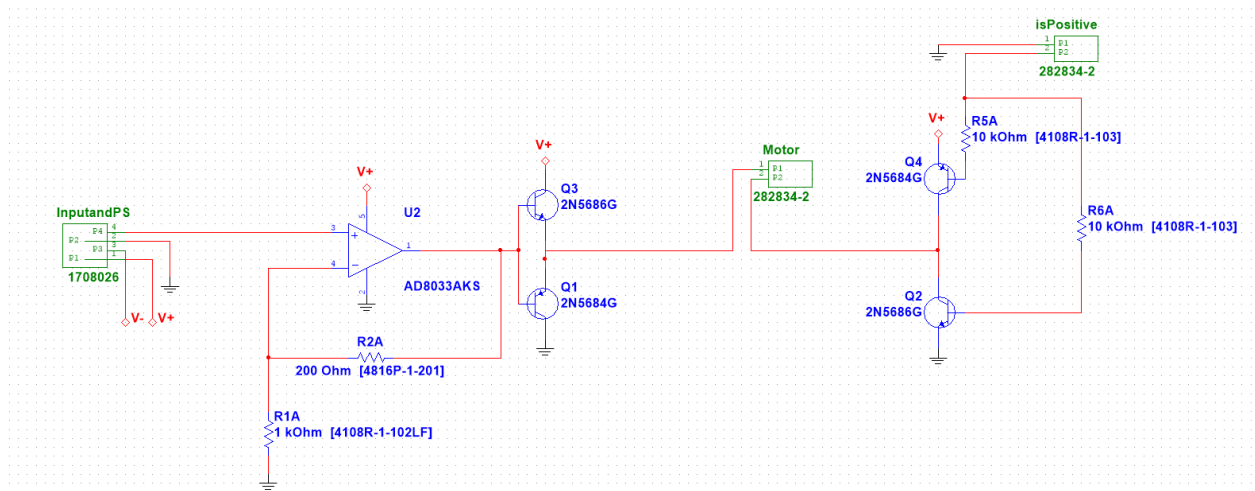


Figure 21: H-Bridge motor driver circuit ready for import to Ultiboard

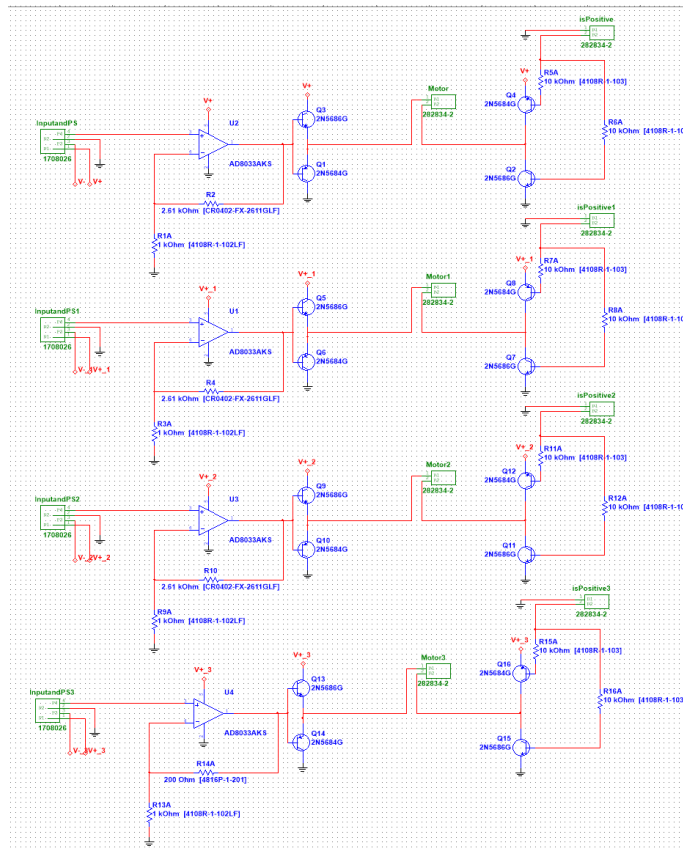


Figure 22: H-Bridge motor driver circuit for DCX10L ready for import to Ultiboard

The final PCB described in the beginning of this section is shown below in Fig. 24. It is clear that there are 4 distinct motor driver circuits (through the silver BJT casings) stacked vertically with the inputs and outputs located at opposite ends. Trace widths of 10mils were used, higher than the industry standard and enough for to provide for the high-current traces. As evident from the PCB, the design is efficient overall, minimizing unnecessary trace lengths.

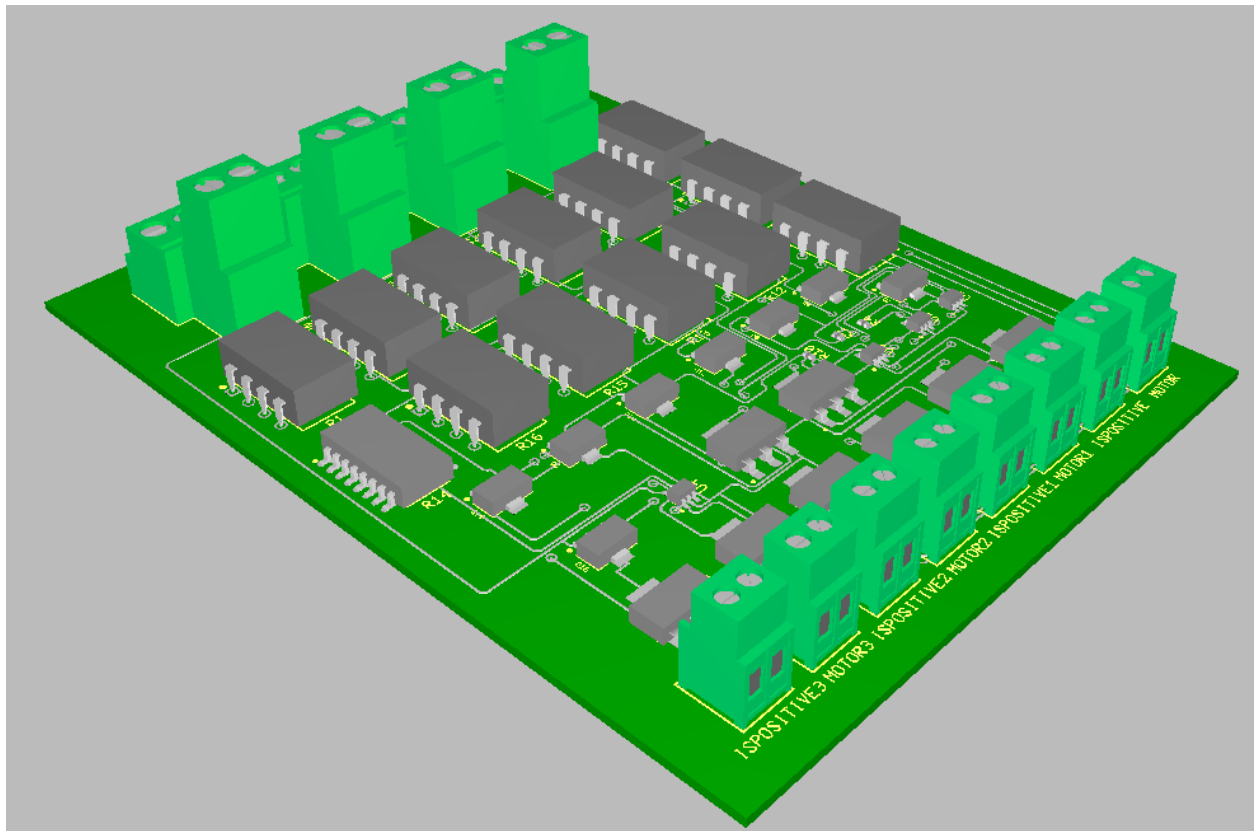
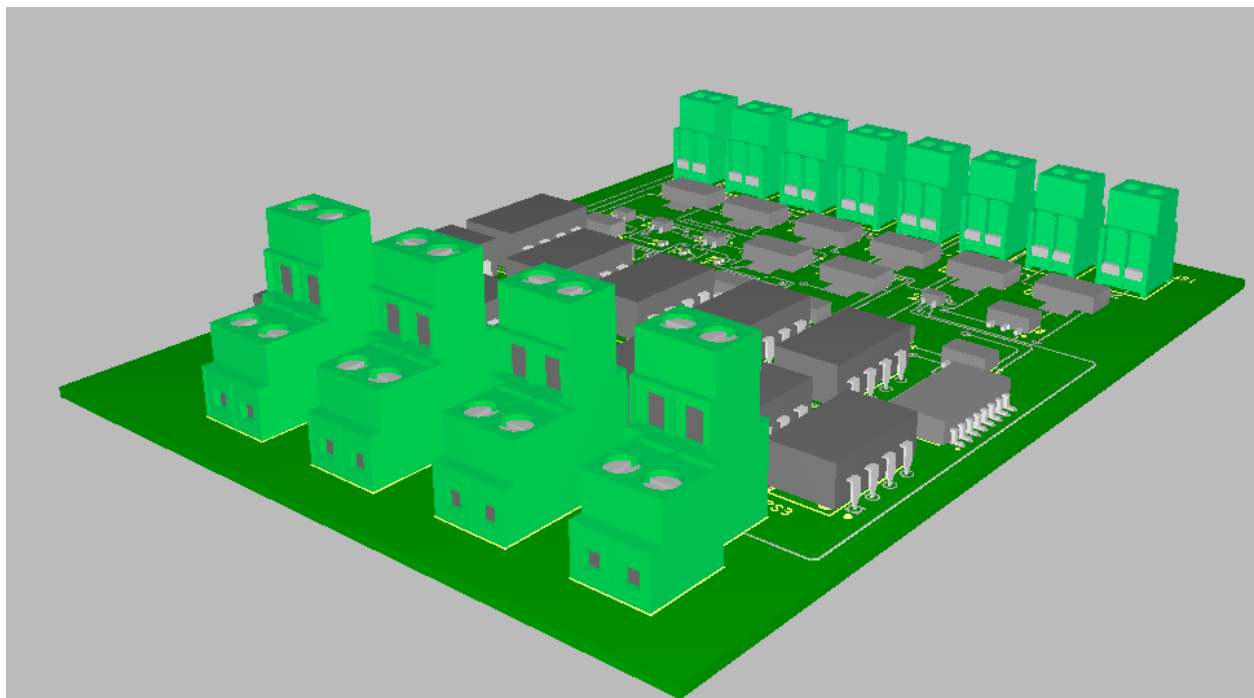
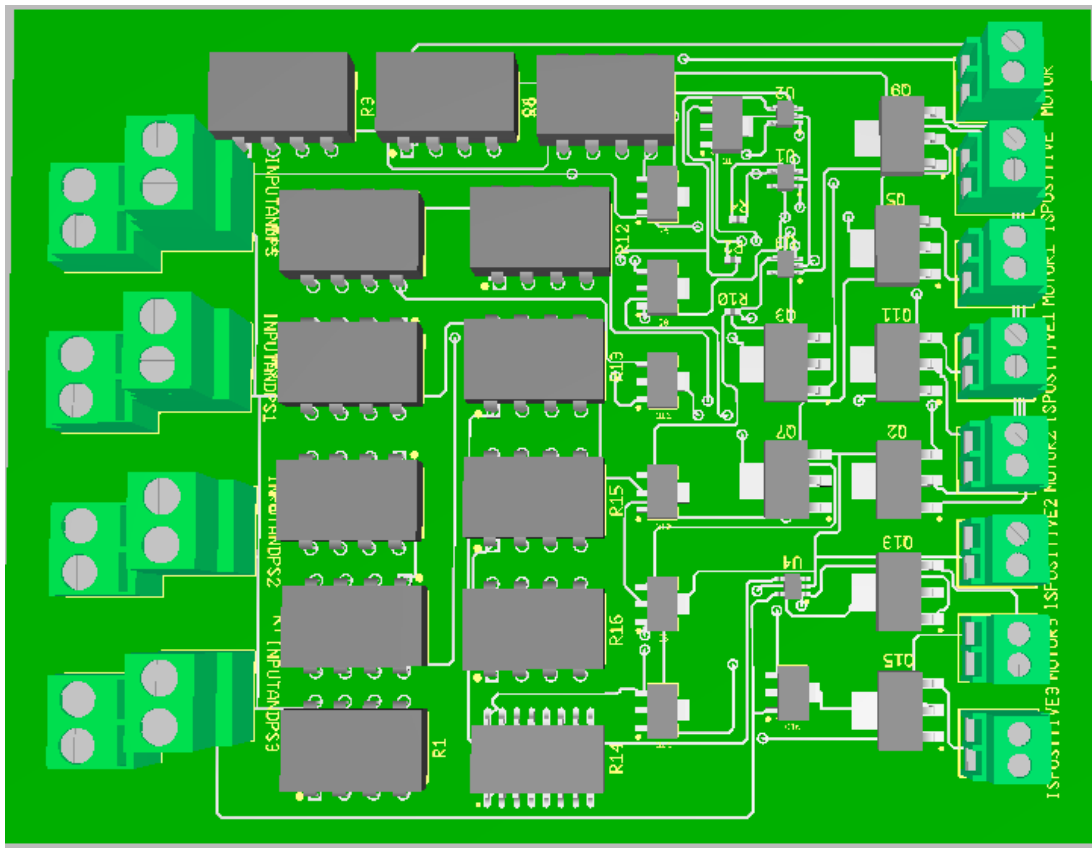


Figure 23: Circuit Design in Multisim with all motor drivers in one





The detailed wiring diagram of the circuit shown above in Fig. 24 is shown in Fig. 25 below.

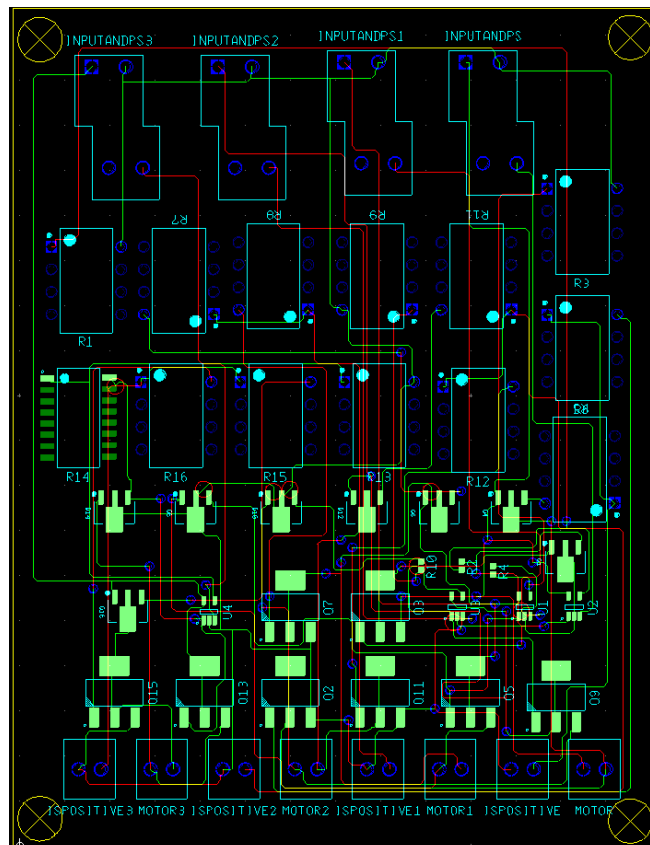


Figure 24: 3D Preview of overall circuit in Ultiboard

F) Cost

A table of the costs associated with this circuit is shown below.

The sum total of the components used in the design is: \$442.04 CAD or \$352.55 USD.

References

[1] <https://www.digikey.ca/en/products/detail/phoenix-contact/1708026/267545>

[2] <https://www.digikey.ca/en/products/detail/analog-devices-inc/AD8033AKSZ-REEL7/671047>

[3] <https://www.digikey.ca/en/products/detail/bourns-inc/CR0402-FX-2611GLF/4697994?s=N4lgTCBcDaIMICUAMAWJYC0AxAGhsAbAIXEDiAMliALoC%2BQA>

[4] <https://www.digikey.ca/en/products/detail/bourns-inc/4108R-1-102LF/3786428?s=N4lgTCBcDaICwEYAMAOASgWgVpYAYiALoC%2BQA>

[5] <https://www.digikey.ca/en/products/detail/on-semiconductor/2N5686G/918245?s=N4lgTCBcDa4HIFYBsAOJBxEBdAvkA>

[6] <https://www.digikey.ca/en/products/detail/on-semiconductor/2N5684G/918244?s=N4lgTCBcDa4HIFYBsAOALAcRAXQL5A>

[7] <https://www.digikey.ca/en/products/detail/te-connectivity-amp-connectors/282834-2/1150135>

[8] <https://www.digikey.ca/en/products/detail/bourns-inc/4108R-1-103/3786429>

[9] <https://www.digikey.ca/en/products/detail/bourns-inc/4816P-1-201/2566213>

[6] <https://www.maxongroup.com/maxon/view/product/sensor/encoder/Magnetische-Encoder/ENX10EASY/ENX10EASY06>

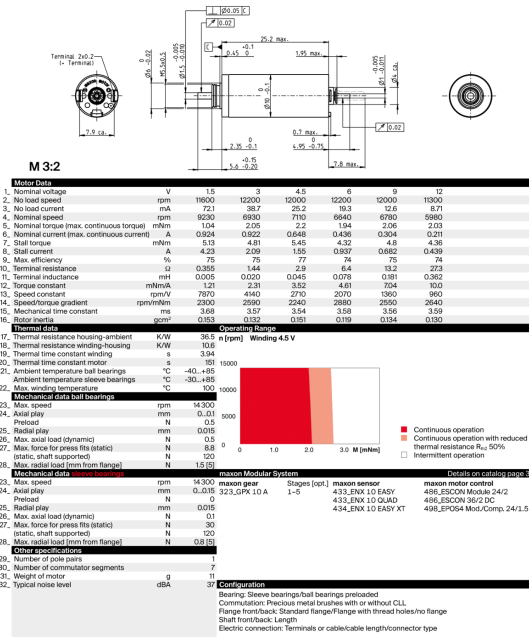
Appendix

Appendix A: Motor Data-sheets

DCX 10 L Precious Metal Brushes
DC motor Ø10 mm

DCX

Key Data: 1.5/3 W, 2.2 mNm, 14300 rpm



xdrives.maxongroup.com

78 maxon DC motor

June 2020 edition / provisional data / subject to change

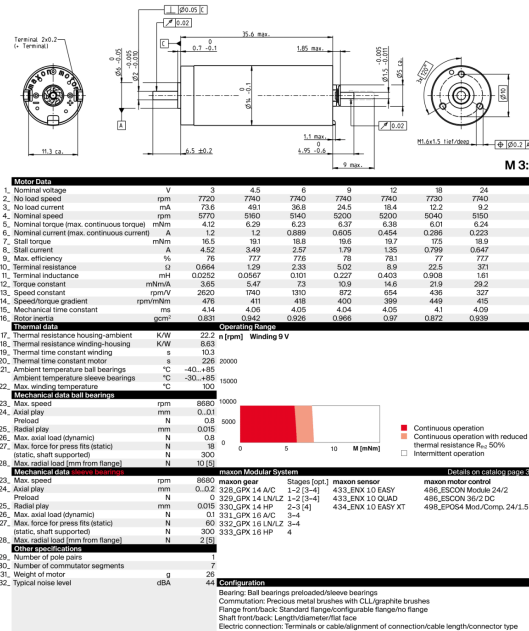
Figure 25: Wiring diagram of overall circuit in Ultiboard

DCX 14 L Precious Metal Brushes
DC motor Ø14 mm

Key Data: 3/5 W, 6.3 mNm, 8680 rpm



DCX



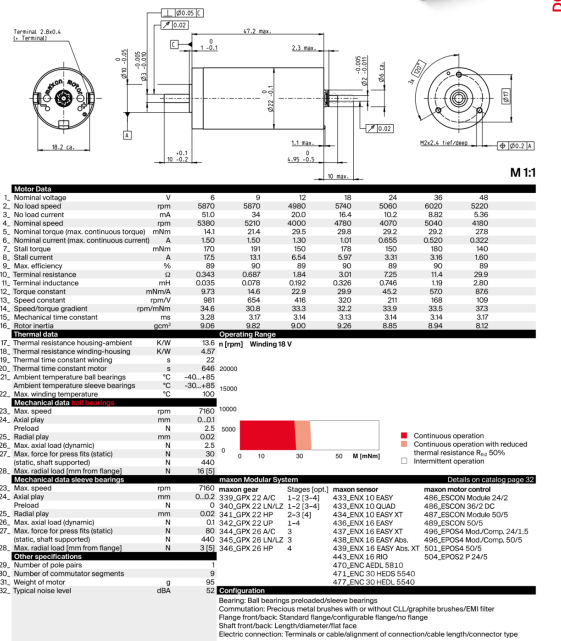
xdrives.maxongroup.com

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maxon DC motor 81

DCX 22 L Precious Metal Brushes DC motor Ø22 mm

Key Data: 11/20 W, 29.8 mNm, 7160 rpm



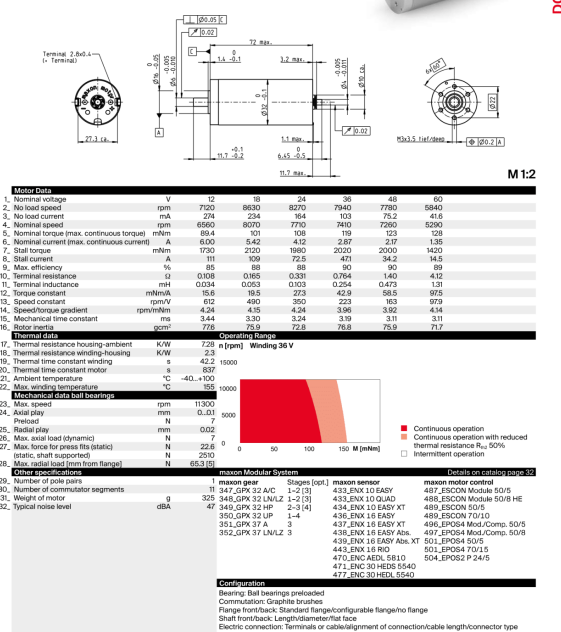
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maxon DC motor 91

DCX 32 L Graphite Brushes DC motor Ø32 mm

Key Data: 70/110 W, 128 mNm, 11300 rpm



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maxon DC motor 95

Extracted Calculation Tables

Table 1

Is Negative	Vsource (Arduino)	Motor
1	0V	Turns Left
1	5V	Off
0	0V	Off
0	5V	Turns Right

Table 2

R (Ω)	0.1650
L (H)	5.3000e-05
No Load Current (A)	0.2340
Torque Constant (Nm/A)	0.0195
No Load Speed (rad/s)	903.7315
Speed Constant (rpm/V)	490
J (Kgm ²)	7.5900e-06
B (Nm/rad/s)	5.0491e-06
Km (Nm/A)	0.0195
Kb (Vs/rad)	0.0195

Table 3

R (Ω)	3.0100
L (H)	3.2600e-04
No Load Current (A)	0.0164
Torque Constant (Nm/A)	0.0299
No Load Speed (rad/s)	601.0914
Speed Constant (rpm/V)	320
J (Kgm ²)	9.2600e-07
B (Nm/rad/s)	8.1578e-07
Km (Nm/A)	0.0299
Kb (Vs/rad)	0.0298

Table 4

R (Ω)	22.5000
L (H)	9.0800e-04
No Load Current (A)	0.0122
Torque Constant (Nm/A)	0.0219
No Load Speed (rad/s)	809.4837
Speed Constant (rpm/V)	436

R (Ω)	22.5000
J (Kgm ²)	8.7200e-08
B (Nm/rad/s)	3.3006e-07
Km (Nm/A)	0.0219
Kb (Vs/rad)	0.0219

Table 5

R (Ω)	6.4000
L (H)	7.8000e-05
No Load Current (A)	0.0193
Torque Constant (Nm/A)	0.0046
No Load Speed (rad/s)	903.7315
Speed Constant (rpm/V)	2070
J (Kgm ²)	1.1900e-08
B (Nm/rad/s)	6.9642e-08
Km (Nm/A)	0.0046
Kb (Vs/rad)	0.0046

Table 6

Item	Count	Unit Price	Provider	URL
1708026	4	5.13 CAD	DigiKey	[1]
AD8033AKSZ-REEL7	4	4.45 CAD	DigiKey	[2]
CR0402-FX-2611GLF	4	0.15 CAD	DigiKey	[3]
4108R-1-102LF	3	1.70 CAD	DigiKey	[4]
2N5686G	8	23.88 CAD	DigiKey	[5]
2N5684G	8	21.7 CAD	DigiKey	[6]
282834-2	8	1.73 CAD	DigiKey	[7]
4108R-1-103	8	2.36 CAD	DigiKey	[8]
4816P-1-201	1	0.66 CAD	DigiKey	[9]